Water diodes and spark gaps in bridges

What useful circuits can be built from aluminium water cells, polarity spark gaps in water and air spark gaps? — CLAUDE.md §3g, §3h

kronspice simulation report · 2026-11-07

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Summary

The companion example *sparkgap_diode* built two one-way devices from water: the **water cell** (aluminium against lead or iron in a phosphate solution, the 1890s rectifier of Pollak and Nodon, one-way up to about 140 V) and the **polarity spark gap** (a point–plane gap in water that breaks down at a lower voltage with the point positive). This example combines them, with plain air spark gaps, into circuits for six jobs, each with an energy ledger.

- **A. DC supply:** four lead–aluminium cells give 87.9 V DC from 100 V peak, 85 % efficient; kronspice gives 87.96 V for the same circuit (−0.01 %).
- **B. Surge protection:** under a 4 kV, 1.2/50 µs surge the largest bridge-cell voltage falls from 221 V (unprotected, above the 140 V spark-over) to 117 V with an air-gap arrester and an aluminium–aluminium clamp.
- **C. Spark pulser:** only an output gap fired at each mains zero crossing works: 120 pulses/s, 505 A peak, 1.13 J per pulse, 50.3 % of mains power to the load. Self-breaking gaps reach 0.6 % and 1.8 %.
- **D. Voltage doubler:** a Greinacher doubler needs two cells per position: 174 V DC at 85 % (one cell: 151 V, 17 %, sparks over).
- **E. Varactor with memory:** the oxide film, set by its forming voltage (10 V to 140 V), tunes a 1 mH tank from 643 Hz to 2,405 Hz, matching $1/(2\pi\sqrt{LC})$ to 1 Hz.
- **F. Frequency doubler:** 60 Hz in, 120 Hz out; 95 % of the output's AC power is at 120 Hz with cells, 89 % with water spark gaps.
- **No energy is added** by cells or gaps: every joule at a load comes from the mains or the surge source.

How it is solved

Circuits with only cells run in kronspice (D, E and the twin of A). A spark gap is a switch with memory: it strikes at its breakdown voltage and stays on while its arc carries current, which kronspice's static elements cannot hold. Circuits with gaps therefore run in the example's own small nodal solver: backward Euler for capacitors and inductors, Newton on the cells' curves, gap states updated after each step. It is checked against an exact RC discharge and against kronspice on the supply. Its energy ledger splits what capacitors and inductors take in into stored energy and backward Euler's numerical damping, and closes to a millionth of the energy supplied or better.

Spark gaps (declared unless cited):

- Water polarity gap: 3 mm point–plane; breakdown with the point positive from Stygar's water relation (extrapolated to 60 Hz), negative ×3; arc 50 V + 0.5 Ω, out below 0.1 A after 10 µs.
- After an arc the breakdown recovers as $1 - e^{-t/1\ \mathrm{ms}}$, and never below the Paschen minimum of water vapour, 480 V (Škoro et al. 2011).
- Air gaps: Paschen's law for air (Raizer), arc 30 V + 0.2 Ω, never below air's Paschen minimum, about 330 V.

Results

A. Mains DC supply

Four lead–aluminium cells, 100 V peak at 60 Hz, 0.5 Ω source, 2,200 µF, 50 Ω load: **87.9 V DC**, ripple 5.2 V, efficiency 85 %. kronspice gives 87.96 V (difference −0.01 %), which checks the example's own solver. Ledger residual 3.7×10^{-9} J of 74.8 J supplied.

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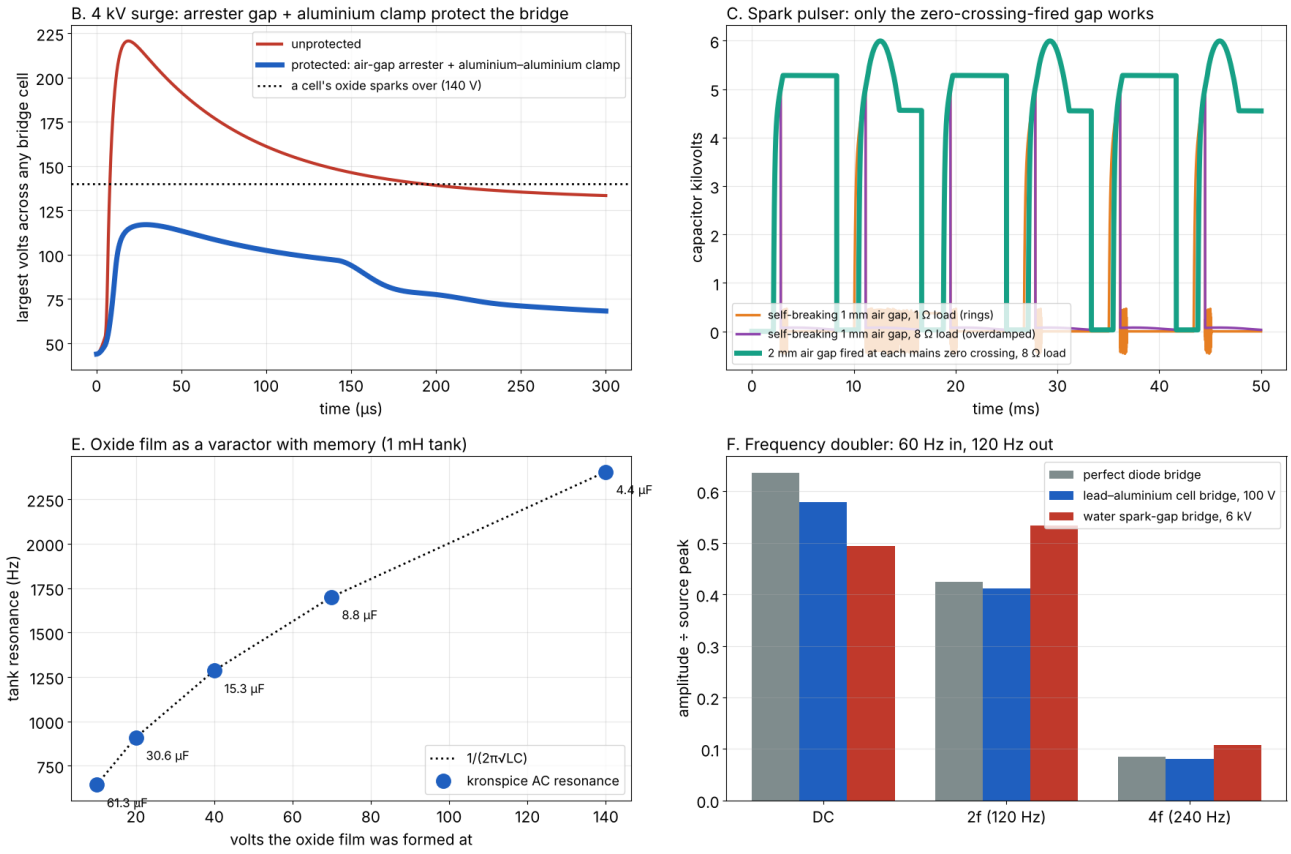


Figure 1. B (top left): largest voltage across any bridge cell during the 4 kV surge, unprotected (red, peak 221 V) and protected (blue, peak 117 V), against the 140 V oxide spark-over (dotted). C (top right): capacitor voltage of the spark pulser for the three output switches; only the zero-crossing-fired gap (green) charges and dumps cleanly every half cycle. E (bottom left): tank resonance against forming voltage, kronspice AC points on the 1/(2π√LC) curve, labelled with the film capacitance. F (bottom right): DC, 120 Hz and 240 Hz amplitudes divided by the source peak for the three bridges.

B. Surge-protected supply

A 4 kV, 1.2/50 µs surge from a 2 Ω source (the IEC 61000-4-5 test-wave shape and impedance; its 8/20 µs current shape is not modelled) arrives on the mains line at the mains peak. Protection has two stages: a 0.2 mm air gap across the line (breaks down at 1,423 V, Paschen's law) takes the bulk of the surge; after 10 µH and 0.5 Ω of line, an aluminium–aluminium water cell across the bridge input blocks both ways up to 140 V and then sparks over, a two-way clamp like a pair of Zener diodes.

	Largest bridge-cell voltage	Extra energy into bridge cells	Arrester	Clamp	DC output peak	Ledger residual
Unprotected	221 V (sparks over at 140 V)	15.42 J	0.0 J (0 strike)	0.000 J	131 V	3×10^{-10} J
Protected	**117 V**	**0.64 J**	**22.9 J (1 strike)**	**0.006 J**	**95 V**	**2×10^{-10} J**

C. Spark pulser

A 6 kV, 60 Hz transformer (1 kΩ) feeds a bridge of four water polarity gaps (polarity ratio 3, the ratio that rectified in *sparkgap_diode*), which charges 0.1 µF. An air gap discharges the capacitor through 1 µH into a load.

Output switch and load	Pulses per s	Peak	To load per pulse	Load / mains	Bridge backward sparks	What happens
Self-breaking 1 mm air gap, 1 Ω (rings)	120	1,200 A	0.73 J	88 W of 13,960 W (0.6 %)	2,564 (15.26 J)	capacitor swings negative, bridge gaps re-strike backwards: the transformer is shorted

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Output switch and load	Pulses per s	Peak	To load per pulse	Load / mains	Bridge backward sparks	What happens
Self-breaking 1 mm air gap, 8 Ω (overdamped)	120	482 A	2.11 J	253 W of 14,081 W (1.8 %)	0 (0.00 J)	the mains keeps the output arc lit (power follow): gap conducting 0.065 s of 0.1 s
2 mm air gap fired at each mains zero crossing, 8 Ω	**120**	**505 A**	**1.13 J**	**135 W of 269 W (50.3 %)**	**2,804 (0.02 J)**	**the transformer cannot follow at its zero; the pulser works**

Firing at a fixed point of the mains cycle is what Tesla-coil builders do with a synchronous rotary spark gap; here the trigger is ideal (declared). In the working design the bridge gaps still make tiny backward sparks, 0.02 J in total, because the transformer and the full capacitor nearly balance. Ledger residuals: 8×10^{-7} J, 8×10^{-7} J and 6×10^{-6} J.

D. Voltage doubler

Greinacher's circuit (a pump capacitor and two diodes; 2,200 µF, 200 Ω) with lead–aluminium cells, 100 V peak, solved in kronspice:

Cells per position	Output	Efficiency	Reverse peak	Ledger mismatch
1	151 V	17 % (sparks over)	160 V	−0.0002
2	**174 V**	**85 %**	**182 V**	**-1×10^{-5}**

Each position sees twice the peak in reverse, so it needs two cells, as in the historical "two or more valves in series for higher pressures".

E. A varactor with memory

The aluminium plate's oxide film is a capacitor whose thickness is set when it is formed: about 1.3 nm per volt (declared, typical barrier anodic alumina). Formed at a higher voltage, the film is thicker and its capacitance smaller. A 1 mH tank across a cell biased in its blocking direction, solved by kronspice AC:

Formed at	Film	Resonance (kronspice)	$1/(2\pi\sqrt{LC})$	Q
10 V	61.30 µF	643 Hz	643 Hz	24
20 V	30.65 µF	909 Hz	909 Hz	33
40 V	15.32 µF	1,285 Hz	1,286 Hz	48
70 V	8.76 µF	1,701 Hz	1,701 Hz	62
140 V	4.38 µF	2,405 Hz	2,405 Hz	87

Biased forward (−5 V) the conducting cell shorts the tank and there is no resonance. **This is not a semiconductor varactor:** its capacitance does not follow the bias and stays fixed below the forming voltage. It is set by the highest voltage the film has seen, a capacitor with memory. Re-forming higher thickens the film at once; thinning it again takes the slow dissolution of the oxide, which is not modelled. Q is low at low forming voltage because the film then leaks more (declared leakage).

F. Frequency doubler

A full bridge's output repeats twice per cycle. Amplitudes divided by the source peak, resistive load:

Bridge	DC	2f (120 Hz)	4f (240 Hz)	Share of AC power at 2f
Perfect diode bridge	0.637	0.424	0.085	95 %
Lead–aluminium cell bridge, 100 V	0.580	0.411	0.081	95 %
Water spark-gap bridge, 6 kV	0.495	0.534	0.108	89 %

A filter tuned to 120 Hz picks out the doubled frequency. The spark-gap bridge's sudden jumps put more into 120 Hz and the higher harmonics than a smooth bridge.

Ledger and validation

- Every run has an energy ledger; the gap solver's ledger closes to a millionth of the energy supplied or better (A: 3.7×10^{-9} J of 74.8 J; B: 2–3×10^{-10} J; C: ≤ 6×10^{-6} J).

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- Twin: the example's solver and kronspice agree on the DC supply to 0.01 %; the solver is also checked against an exact RC discharge.
- Varactor resonances from kronspice AC agree with $1/(2\pi\sqrt{LC})$ to within 1 Hz.
- Tests in *main_test.go*: model match, solver checks, the twin, and each job's result.

Limits of the model

- The water gap's breakdown extrapolates Stygar's pulsed-power relation to 60 Hz; the negative/positive polarity ratio, arc constants and 1 ms recovery are declared.
- The surge's 8/20 µs current shape is not modelled; the zero-crossing trigger in C is ideal.
- The oxide's growth rate (1.3 nm/V) and leakage are declared; slow dissolution of the film is not modelled.

Reproducing the results

```
go run ./examples/sparkgap_diodebridge -viz examples/sparkgap_diodebridge/viz   # about 20 seconds
python3 examples/sparkgap_diodebridge/viz/render.py
go test ./examples/sparkgap_diodebridge
```

Sources: Encyclopaedia Britannica, 11th ed. (1911), vol. 27, "Transformers"; H. Greinacher, Z. Phys. 4, 195 (1921); IEC 61000-4-5; Yu. P. Raizer, Gas Discharge Physics (1991); W. A. Stygar et al., Phys. Rev. ST Accel. Beams 9, 070401 (2006); Škoro et al., Phys. Rev. E 84, 055401 (2011).

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